

Module 4 — Read 1

Causal questions, study design, and confounding

Module 4, Read 1: Causal questions, confounding, and the RCT

Project: Project 4 — Designing Fair Comparisons · **Preview:** Read 0 · **Prerequisite:** Module 3 · **Next:** Quiz A (Def & Notation), then Read 2.

This reading is your **self-paced textbook** for Module 4, Part 1. Work one **section** at a time; use **Check yourself** boxes before **Quiz A**. You will learn to distinguish **observational studies** from **randomized experiments**, identify **confounders**, and understand what p-values can and cannot establish about causation.

Learning outcomes

Work through both sections below in order. Each section ends with **Check yourself** questions.

After this reading you should be able to:

Section 1: Causal questions and study design

Inference for two-group comparisons — causal questions and study design

- Compare **randomized controlled trials** and **observational** designs when the goal is a causal claim about a treatment or exposure.
- Explain how participants end up in comparison groups in an **RCT** versus an **observational study**.
- State what a p -value does and does **not** address for a difference between two groups.

Section 2: Confounding and the RCT

Confounding

- Define a **confounder** in words and recognize plausible confounders in a stated observational comparison.
- Explain how a confounder can **create or distort** an association between an explanatory variable and a response.
- Describe **subclassification** / **stratification** when a confounder is measured.

Section 1: Causal questions and study design

Causal questions and study design

Project 4 connection

Project 4 asks you to design a **fair comparison** between two groups. This section gives you the vocabulary: **EV**, **RV**, **RCT vs observational**, and what a p -value can and cannot tell you about causation.

Observational studies and confounding

This lecture demonstrate how to do a fair comparison of two groups using the **observational** study design: the groups exist before you analyze them (which university, sleep habits in the wild, nightlight use, etc.). Lecture 27 gave you tables and plots for two categorical variables; here we connect those tools to **causal** questions, **three** competing explanations for any difference, and **Causal Directed acyclic graphs (DAGs)** and **subclassification** tell you what to measure and how to compare fairly within levels of a confounder when that is possible.

Learning objectives

- Recognize an **observational** study design and list three explanations for a observed difference between two groups (**causation**, **chance**, **confounding**).
- Define a **confounding variable** both in words and in a DAG.
- Use the method of **subclassification** to adjust a group comparison for a confounding variable to get a fairer comparison.
- Perform a large-sample version of a hypothesis test or confidence interval for **two proportions** using `prop.test()` to eliminate chance as an explanation for the observed difference and estimate the causal effect of group membership on the effect of interest.

Causal questions, explanatory and response variables

A **causal** question asks about the **effect** of something. Name:

Role	Other names	Meaning
Response	outcome, dependent	The effect you want to compare.
Explanatory	treatment, exposure	The potential cause or group label.

When writing a research question that is causal, it is best practice to:

1. Define variables precisely: State how each variable is measured and any cutoffs or rules (e.g. pass/fail, deprived vs unrestricted sleep, income in dollars). Vague labels make design and analysis impossible.
2. Identify the observational units: The unit is the entity that gets a value for each variable—often one person at one time point for this unit (e.g. one student in one calculus term). Your dataset is usually one row per observational unit.
3. If the response is numerical, you typically summarize comparison of two groups with means (or medians) and a Welch two-sample t -test** or randomization test on the difference in means (Lecture 29, Lecture 30). If the

response is binary (or you reduce an outcome to two categories), you compare proportions or rates and use `prop.test` / `chisq.test` or a chi-square shuffle applet (this lecture, Lecture 30).

Note: **Project 4** will asks you to **plan** a study to answer a causal question from one of your classamtes.

Observational studies

In an **observational study**, researchers **record** which group each unit is in; they do **not randomly assign** the explanatory variable. Examples: students **choose** a university; parents **choose** nightlight use for their children; people decide what to eat for lunch.

A **difference** in sample proportions or means from an observational study can reflect:

1. A **Causal effect** of the explanatory variable on the response.
2. **Chance** (random variation—even numbers of heads from fair coins differ in 100 flips).
3. **Confounding**: a third variable influenes both the explanatory and response variables and mixes up relationship between the these two variables,making a fair comparison difficult (**internal validity** is weak for causation until you address it).

Lecture 29 covers **RCTs**, where **random assignment** is used to **balance** confounders **in expectation** and support **causal** conclusions about the assigned treatment (with the usual caveats).

Students often confuse **random assignment** (group assignment within the sample) with **random sampling** (selection of individuals from a population): both are useful, but for different goals. Random assignment is used to adjust for confounding between the groups, making a comparison between the groups more fair. Random sampling is used to elimated bias when generalizing from the study data back to a larger population.

Which university is better?

Suppose a certain state has two polytechnic universities and the Chancellor's Office is trying to decide which one is doing a better job teaching **calculus** to **incoming freshmen**.

*This example is a modification of a story in Pearl, Judea, and Dana Mackenzie, **The Book of Why: The New Science of Cause and Effect** (Basic Books, 2018).*

Crude pass rates

Both universities have a calculus class of **200** students. The number (and percentage) of students who **passed** is:

	Passed calculus
Timber Ridge	150/200 (75%)
Wild Horse	154/200 (77%)

Wild Horse has **more students who passed** in this cohort (154 vs 150).

More students passed—but is it a fair comparison?

The table above compares observational groups: students chose (or landed at) a university; the Chancellor did not randomly assign them. A higher crude pass rate at Wild Horse does not by itself answer the causal question: *Does attending Wild Horse cause a higher chance of passing than attending Timber Ridge?* We still have to worry about chance and confounding as possible explanations.

If we know who took AP Calculus in high school

	AP calculus	No AP calculus	Combined
Timber Ridge	45/50 (90%)	105/150 (70%)	150/200 (75%)
Wild Horse	136/170 (80%)	18/30 (60%)	154/200 (77%)

Each cell shows **number who passed / number of students** in that cell.

Read down each column (holding preparation fixed):

- Among students **without** AP calculus in high school, **Timber Ridge** has the higher pass rate (70% vs 60%).
- Among students **with** AP calculus, **Timber Ridge** again has the higher pass rate (90% vs 80%).

So **within** each preparation level, Timber Ridge does better.

If we know who used the campus tutoring center

	Used tutoring center on campus	Did not use tutoring center	Combined
Timber Ridge	86/120 (71%)	64/80 (80%)	150/200 (75%)
Wild Horse	65/90 (72%)	89/110 (81%)	154/200 (77%)

Within “did not use tutoring,” Wild Horse’s pass rate is slightly higher (81% vs 80%). **Within** “used tutoring,” Wild Horse is also slightly higher (72% vs 71%). But does this breakdown make a fair comparison between the two universities?

Should we “adjust” for tutoring?

We should not treat the tutoring breakdown as the right way to fairly compare the causal effect of **which university** a student attends when tutoring centers are part of what the university provides. If we do breakdown the rates by tutoring or not, we’d remove part of how the university affects success: the tutoring center is on the causal pathway from university to passing, not merely a pre-enrollment mix of students like AP preparation in high school.

A tool for thinking clearly about when to act

When people see tables like these (even faculty audiences), votes are often split on “which school is better.” Causal DAGs are an easy way to see that AP calculus behaves like a confounder you may want to condition on for a fairer comparison of universities, while tutoring center use is not the same kind of adjustment target if tutoring is part of the university’s effect.

Confounding variable (definition)

A **confounder** (for the effect of explanatory variable X on response Y) is a variable that influences both X and Y .

Example: **AP calculus** in the story above confounds **university** and **passing** because it influences **both** where students enroll and how likely they are to pass.

Example: **Nightlight and myopia**: genetics and parental myopia may confound nightlight use and child myopia.

Adjustment sets and the backdoor idea

DAG recap. **Nodes** = variables. **Directed edge** $A \rightarrow B$ = “ A is posited to help cause B in this simplified story.” **Acyclic** = no directed cycles.

Backdoor path (informal definition). For the causal effect of **university** U on **passing** P , a **backdoor** path is a non-causal route that “leaks” association into the crude U – P link—e.g. $U \leftarrow A \rightarrow P$ when A is **AP calculus**.

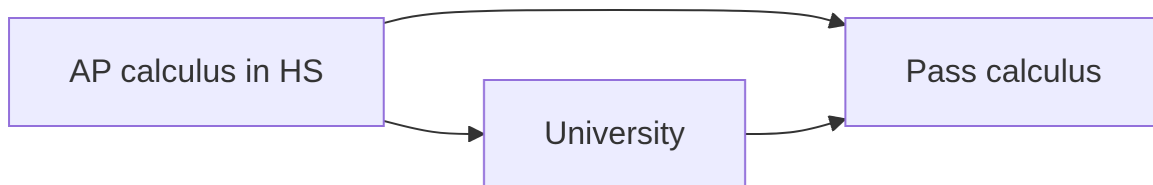
An **adjustment set** (backdoor adjustment, introductory version) is a set of variables you **condition** on (here, mainly **subclassification**: compare universities **within** levels of A) so that, **if** the DAG is correct and other assumptions hold, the **conditional** comparison is **less biased** for the effect of U on P .

The figure below is a **DAGitty** graph for this teaching example (university, preparation, tutoring, passing, and related variables as you drew them in the async course). Use **DAGitty** to reopen the model and highlight **adjustment sets** under your assumptions.

- **AP calculus** (or analogous preparation) typically sits on a **backdoor** path into the crude university–pass association: $U \leftarrow \text{AP} \rightarrow P$. **Conditioning on AP** (the **AP / no AP** subtables) **blocks** that path.
- **Tutoring center use** is **downstream** of **university** on a **path** to **passing** in the story where the center is part of what the university provides. **Conditioning on tutoring** can then **block** part of the **total effect** of university you want to describe when the policy question is the **whole package** at each school.

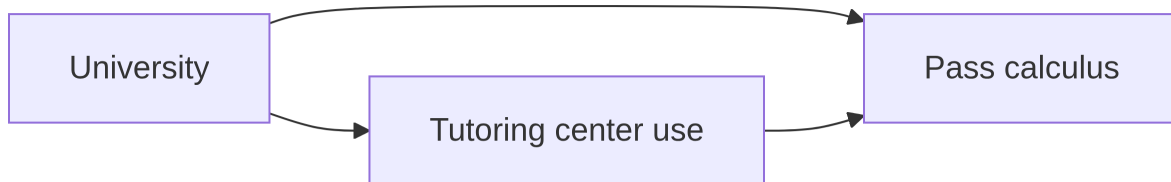
flowchart LR

```
A[AP calculus in HS] --> U[University]
A --> P[Pass calculus]
U --> P
```



flowchart LR

```
U2[University] --> T[Tutoring center use]
U2 --> P2[Pass calculus]
T --> P2
```



If you believe the diagram, which subclassification is appropriate?

Under the **AP** DAG, the appropriate breakdown for comparing universities on pass rates while addressing that backdoor is the AP vs no AP table—not the tutoring table used as if it were the same kind of confounder.

	AP calculus	No AP calculus	Combined
Timber Ridge	45/50 (90%)	105/150 (70%)	150/200 (75%)
Wild Horse	136/170 (80%)	18/30 (60%)	154/200 (77%)

	Used tutoring	Did not use tutoring	Combined
Timber Ridge	86/120 (71%)	64/80 (80%)	150/200 (75%)
Wild Horse	65/90 (72%)	89/110 (81%)	154/200 (77%)

Do you believe the diagram?

Before adjusting, ask: **Did we include every variable that needs to be in the story**, that is, every common cause or fork that links **university**, **passing**, and other measured variables? **Residual confounding** from **unmeasured** variables is always possible.

Finding an adjustment set: by hand or with software

You can **trace** backdoor paths on the board and list a set of variables to **condition** on, or use **DAGitty** under an **assumed** graph to suggest **adjustment sets**.

So which university is better?

To make a **fairer** comparison of **calculus passage rates** across universities—if you accept a DAG like the one above (and usual positivity / correct functional-form ideas we skip in this intro), we would want data on **student motivation**, **socioeconomic status (SES)**, and **AP calculus** (and **university** and **pass/fail**), then compare universities within levels of the variables in the adjustment set you defend. **AP alone** is already enough to change the story in this toy example; thinking about the data we don't have on **motivation** and **SES** reminds us that real policy questions usually need datasets with more variables.

Ruling out chance (not confounding)

Hypothesis tests and p-values address explanation (2) only: could the data look like this by chance alone? That is, what would we expect to see if there were no causation or confounding model? A p-value does **not**, by itself, choose us choose between (1) causation and (3) confounding.

A. Randomization-based (nonparametric) inference — Rossman–Chance

From the [applet collection](#):

- **Randomization test: two groups, quantitative** — shuffle **group labels**; build a null distribution of the **difference in means** (or medians if the applet allows). **No normality** assumption for the **randomization p-value**; the logic is “if treatment labels were arbitrary, how often would we see a gap this big?”
- **Randomization: categorical / chi-square shuffle** — shuffles consistent with **testing no association** in a table (see also [Dolphin study](#)).
- **Two-way table simulation** — builds intuition for **counts** under independence.
- **Guess the p-value** — links **tail areas** to strength of evidence.

B. Large-sample tests for two proportions — theory-based p-values in R

For **binary** outcomes (pass/fail, infected or not) compared across **two** groups, common large-sample tools are `prop.test()` (equivalent 2×2 logic to a z-test for two proportions) and `chisq.test()` on the **contingency table** of counts. **Conditions**:

1. **Independent** observations.
2. **Large expected counts** under the null: typically **all expected cell counts** ≥ 5 for Pearson’s chi-squared; inspect `$expected`. If violated, use `simulate.p.value = TRUE`

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

uni_crude <- tibble(
  university = rep(c("Timber Ridge", "Wild Horse"), c(200, 200)),
  passed = c(rep(c(TRUE, FALSE), c(150, 50)), rep(c(TRUE, FALSE), c(154, 46)))
)

tab_uni <- table(uni_crude$university, uni_crude$passed)
tab_uni
```

	FALSE	TRUE
Timber Ridge	50	150
Wild Horse	46	154

Two-sample proportion test:

```
prop.test(x = c(150, 154), n = c(200, 200), correct = FALSE)
```

2-sample test for equality of proportions without continuity correction

data: c(150, 154) out of c(200, 200)
X-squared = 0.2193, df = 1, p-value = 0.6396
alternative hypothesis: two.sided
95 percent confidence interval:


```

-0.10368381  0.06368381
sample estimates:
prop 1 prop 2
  0.75   0.77

```

Chi-squared test of independence (check expected counts first):

```

cq <- chisq.test(tab_uni)
cq$expected

```

	FALSE	TRUE
Timber Ridge	48	152
Wild Horse	48	152

```

cq

```

Pearson's Chi-squared test with Yates' continuity correction

```

data:  tab_uni
X-squared = 0.12336, df = 1, p-value = 0.7254

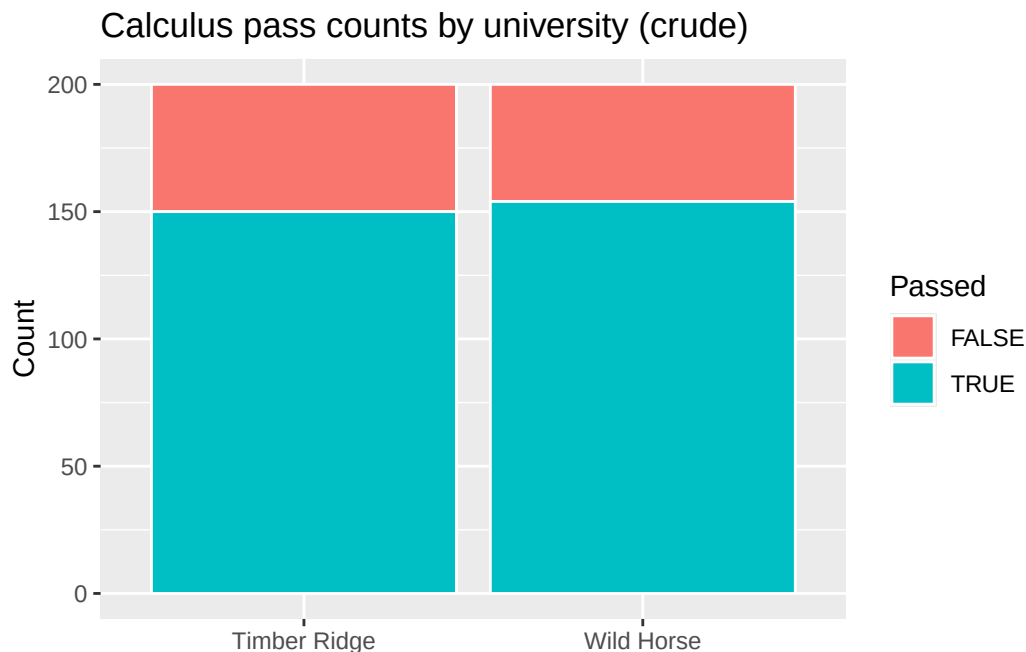
```

Stacked bar chart (Lecture 27)

```

uni_crude |>
  count(university, passed) |>
  ggplot(aes(x = university, y = n, fill = passed)) +
  geom_col(color = "white") +
  labs(
    title = "Calculus pass counts by university (crude)",
    x = NULL,
    y = "Count",
    fill = "Passed"
  )

```



Interpretation: A large or small p-value for speaks to only whether this data is consistent with a **by chance explanation**, not to whether attending Wild Horse causes a higher pass rate.

::: {.callout-tip icon=false} ## Check yourself — Causal questions and study design

1. In a study comparing smokers and non-smokers on lung cancer rates, which design is this — RCT or observational?
2. What is an **explanatory variable** (EV) and a **response variable** (RV)? Give an example.
3. A p-value of 0.03 is found for a group difference in an observational study. Does this prove the treatment caused the difference? Explain.
4. Sketch a simple DAG with one confounder C that affects both EV X and outcome Y .

:::

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1. **Observational** — researchers cannot randomly assign people to smoke. Group membership is self-selected.
2. **EV:** the variable that explains or predicts (e.g. treatment group). **RV:** the outcome of interest (e.g. recovery status). Directionality reflects the research question.
3. **No.** A p-value only addresses whether the difference is surprising *if there were no true difference*. Confounders in an observational study can still explain the association causally.
4. $C \rightarrow X$ and $C \rightarrow Y$. Without controlling for C , the $X \rightarrow Y$ path includes confounded association.

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Answers — Causal questions and study design

1. **Observational** — researchers cannot randomly assign people to smoke. Group membership is self-selected.
2. **EV:** the variable that explains or predicts (e.g. treatment group). **RV:** the outcome of interest (e.g. recovery status). Directionality reflects the research question.
3. **No.** A p-value only addresses whether the difference is surprising *if there were no true difference*. Confounders in an observational study can still explain the association causally.
4. $C \rightarrow X$ and $C \rightarrow Y$. Without controlling for C , the $X \rightarrow Y$ path includes confounded association.

:::

Section 2: Confounding and the RCT

Confounding and the randomized controlled trial

Bridge from Section 1

Section 1 introduced **observational vs experimental** designs. Here we name the mechanism that makes observational comparisons tricky: the **confounder**. The **RCT** is the gold standard precisely because random assignment breaks confounding paths.

Random assignment and what it buys you

This lecture focuses on **randomized controlled trials (RCTs)**. **Random assignment** of the group membership is the design feature that **replaces** the main threat to internal validity, confounding.

Learning objectives

- Describe an **RCT** for a response and explanatory variable of interest and explain how random assignment could be carried out.
 - List **two** primary explanations for an observed difference in an RCT (**treatment effect, chance**)
 - Distinguish **random assignment** and **random sampling** and say what each
 - For a **numerical** outcome and two groups, compute **Lecture 25** descriptives, plot **boxplots / histograms**, and run **Welch's two-sample t -test** (`t.test(..., var.equal = FALSE)`) after checking **conditions**.
-

RCT definition

A **Randomized Controlled Trial (RCT)** is a study design where the researcher randomly assigns people/units to one of two or more groups. This assigned group membership becomes the explanatory variable and a response variable of interest is measured for each unit.

Internal validity in an RCT

Internal validity = how well the study rules out alternative explanations for an observed difference between the groups within the study.

Under ideal conditions, random assignment balances both measured and unmeasured baseline confounders on average.

Numerical outcome: sleep condition and improvement

Research question. Does **sleep deprivation** change **improvement** in reaction time from a baseline test to a test three days later?

Design (for class). Suppose students were **randomly assigned** to **deprived vs unrestricted** sleep groups before

the follow-up test of reaction times.

Hypotheses (population means). Let μ_{dep} and μ_{unr} be mean **improvement** in the two groups.

- H_0 : $\mu_{\text{dep}} = \mu_{\text{unr}}$
- H_a : $\mu_{\text{dep}} \neq \mu_{\text{unr}}$

Data ($n = 21$)

sleep condition	improvement
unrestricted	-14.7
unrestricted	14.5
deprived	2.2
deprived	21.3
unrestricted	11.5
deprived	13.7
deprived	9.6
deprived	22.4
deprived	21.1
deprived	7.2
deprived	10.0
unrestricted	25.2
unrestricted	-11.2
unrestricted	7.0
unrestricted	14.7
deprived	4.5
unrestricted	45.6
unrestricted	15.6
unrestricted	18.8
unrestricted	12.1
unrestricted	10.3

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {  
  install.packages("tidyverse", repos = "https://cloud.r-project.org")  
}  
  
library(tidyverse)  
  
sleep_rct <- tibble(  
  sleepcondition = c(  
    "unrestricted", "unrestricted", "deprived", "deprived", "unrestricted",  
    "deprived", "deprived", "deprived", "deprived", "deprived",  
    "deprived", "unrestricted", "unrestricted", "unrestricted", "unrestricted",  
    "deprived", "unrestricted", "unrestricted", "unrestricted", "unrestricted",  
    "unrestricted"  
  ),  
  improvement = c(  
    -14.7, 14.5, 2.2, 21.3, 11.5, 13.7, 9.6, 22.4, 21.1, 7.2, 10.0,  
    25.2, -11.2, 7.0, 14.7, 4.5, 45.6, 15.6, 18.8, 12.1, 10.3  
  )  
) |>  
mutate(  
  #
```

```
sleepcondition = factor(sleepcondition, levels = c("deprived", "unrestricted"))
)
```

Descriptive statistics (Lecture 25)

```
sleep_rct |>
  group_by(sleepcondition) |>
  summarize(
    n = n(),
    mean_improvement = mean(improvement),
    sd_improvement = sd(improvement),
    median_improvement = median(improvement),
    iqr_improvement = IQR(improvement)
  )
```

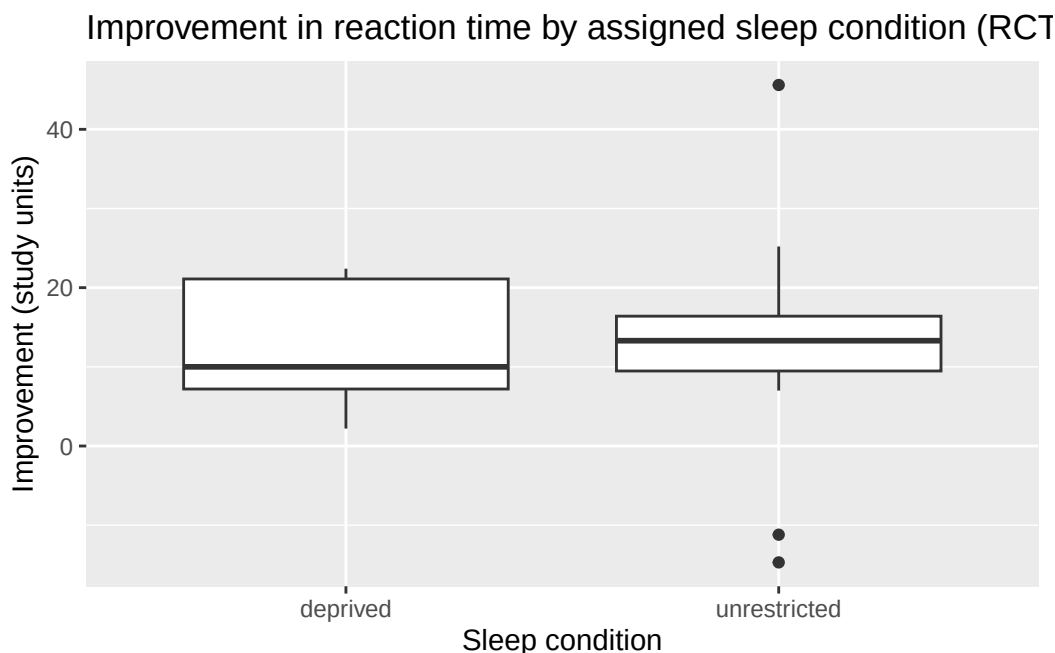
A tibble: 2 x 6

sleepcondition	n	mean_improvement	sd_improvement	median_improvement
<fct>	<int>	<dbl>	<dbl>	<dbl>
1 deprived	9	12.4	7.61	10
2 unrestricted	12	12.4	15.5	13.3

i 1 more variable: iqr_improvement <dbl>

Plots

```
ggplot(sleep_rct, aes(x = sleepcondition, y = improvement)) +
  geom_boxplot() +
  labs(
    title = "Improvement in reaction time by assigned sleep condition (RCT framing)",
    x = "Sleep condition",
    y = "Improvement (study units)"
  )
)
```



Welch two-sample t -test (Lecture 25 recap)

Conditions (practical): Independence between all students in both groups; small sample sizes so we'll have to assume the populations of improvement are approximately normal.

```
t.test(improvement ~ sleepcondition, data = sleep_rct, var.equal = FALSE)
```

Welch Two Sample t-test

```
data: improvement by sleepcondition
t = -0.0010785, df = 16.806, p-value = 0.9992
alternative hypothesis: true difference in means between group deprived and group unrestricted is not equal
95 percent confidence interval:
 -10.88355  10.87244
sample estimates:
mean in group deprived mean in group unrestricted
      12.44444             12.45000
```

Conclusion from RCT. If the design is truly randomized and well executed, a small p -value supports that the mean improvement differs by assigned sleep condition beyond what chance alone would typically produce.

Binary outcome in an RCT:

Story. 200 patients per arm; count **infections** (yes/no).

Hypotheses (population proportions). Let p_V and p_P be the **population** probabilities (or long-run proportions) of infection under **vaccine** and **placebo** assignment in the target setting.

- $H_0: p_V = p_P$ (no difference in infection probability between the two groups).
- $H_a: p_V \neq p_P$ (there is a difference in infection probability between the two groups).

```
set.seed(2901)
vax_trial <- tibble(
  arm = factor(rep(c("Vaccine", "Placebo"), each = 200)),
  infected = c(rbinom(200, 1, 0.08), rbinom(200, 1, 0.14))
)
```

```
tab_v <- table(vax_trial$arm, vax_trial$infected)
tab_v
```

	0	1
Placebo	174	26
Vaccine	178	22

```
infected_vaccine <- sum(vax_trial$infected[vax_trial$arm == "Vaccine"])
n_vaccine <- sum(vax_trial$arm == "Vaccine")
infected_placebo <- sum(vax_trial$infected[vax_trial$arm == "Placebo"])
n_placebo <- sum(vax_trial$arm == "Placebo")

prop.test(x = c(infected_vaccine, infected_placebo), n = c(n_vaccine, n_placebo), correct = FALSE)
```

2-sample test for equality of proportions without continuity correction

```
data: c(infected_vaccine, infected_placebo) out of c(n_vaccine, n_placebo)
X-squared = 0.37879, df = 1, p-value = 0.5383
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.08366113  0.04366113
sample estimates:
prop 1 prop 2
 0.11  0.13
```

```
chisq.test(tab_v)$expected
```

```
      0  1
Placebo 176 24
Vaccine 176 24
```

```
chisq.test(tab_v)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: tab_v
X-squared = 0.21307, df = 1, p-value = 0.6444
```

Here **random assignment** makes a **low** infection rate in the vaccine arm **more** plausibly a **treatment effect** than the same table would be in an **observational** comparison.

When an RCT is impossible or unethical

Some groups cannot be randomly assigned (e.g. long-term **smoking** in teens).

Check yourself — Confounding and the RCT

1. Define **confounder** in one sentence.
2. In a study of coffee and heart disease, age might be a confounder. Why?
3. What does **random assignment** accomplish in an RCT that an observational study cannot guarantee?
4. If a confounder is **measured**, what technique can partially account for it?

Answers — Confounding and the RCT

1. A **confounder** is a variable associated with both the explanatory variable and the response that can create or distort the apparent association between them.
2. Older people drink more coffee **and** are more likely to have heart disease — so age distorts the coffee–heart disease association.
3. Random assignment distributes known **and unknown** confounders approximately equally across groups on average, so group differences are attributable to the treatment.
4. **Subclassification / stratification** — compare treatment groups within subgroups defined by the confounder value.

i Optional enrichment (not required for Project 4)

- **Lecture 31** — matched pairs and McNemar’s test for paired binary data
- **Lecture 34** — work session and simulation walkthrough (15-week path)